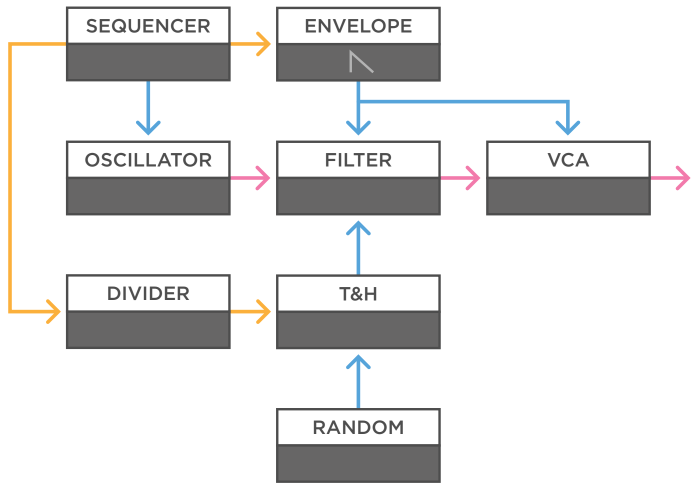
Track & Hold

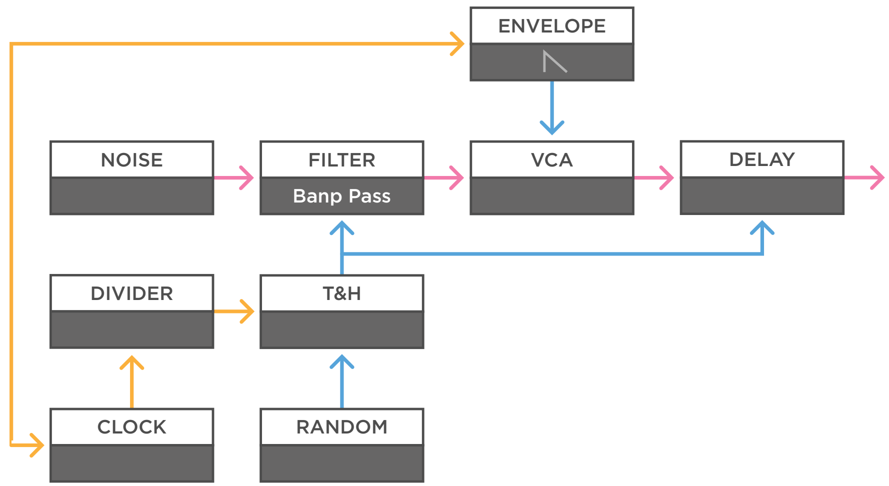
Patch-tips

Audio
Control Voltage
Trigger / Gate

MONOTRAIL TECH TALK



(a)



(b)

a - Sending a smooth random voltage to a track & hold module will give you the possibility to freeze the random voltage at any time. This can work great in all sorts of patches. For example, when you make a simple voice with an oscillator, filter, and VCA, use a sequencer to create a melody, and trigger an envelope modulating the filter and VCA. In this case you can use the sequencer to drive a clock divider, and have a slower division gate the track & hold. Send the outcome of the track & hold to the filter in the voice.

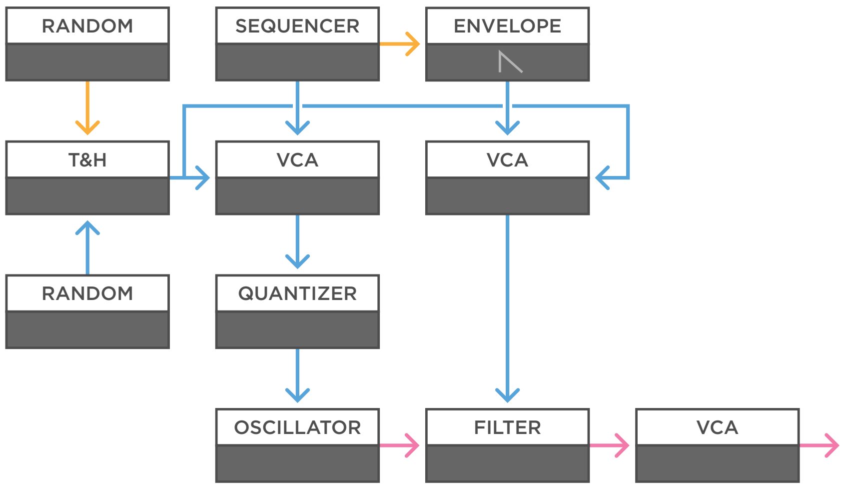
b - If you want to create some interesting percussion, you can send a noise source to a band pass filter, a VCA, and a delay module. Then add a track & hold module that you feed a medium speed random voltage, as well as a division from a clock divider, driven by a modular clock. You can use the same clock to trigger an envelope with a fast attack and short decay to open the VCA. And if you want even more dynamics, you can multiply the output of the track & hold, and send that to modulate the delay time as well.

Track & Hold

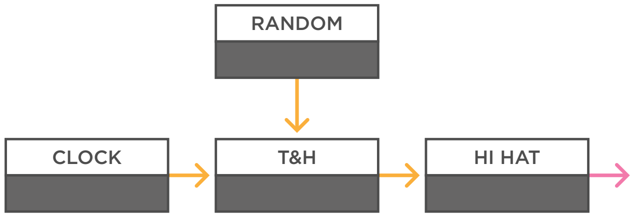
Patch-tips

Audio
Control Voltage
Trigger / Gate

MONOTRAIL TECH TALK



(a)



(b)

a - In this patch you see a simple voice, again with a sequencer and envelope. But this time both the melodic sequence, and the envelope, are sent to a VCA. After the VCA, the sequenced melody passes on to a quantizer, and then the oscillator. The envelope, passes on to the filter. Add a track & hold module with two random voltages, one to track, the other to create random gates.
Then multiply the output to both the VCAs. Now, when the voltage is low, there is no melodic sequence, and no envelope. However, when the voltage goes high, both the melody, and the modulation to the filter will grow in strength.

b - You can also use track & hold to interrupt a trigger pattern. If you feed a steady clock to the track input, you can let that signal pass on to trigger something like a hi-hat. Now, you can experiment with different gate inputs to hold the signal, and interrupt the pattern. You can use something like an un-synced square wave LFO, but smooth random voltages also work great to create random gated patterns.

Track & Hold

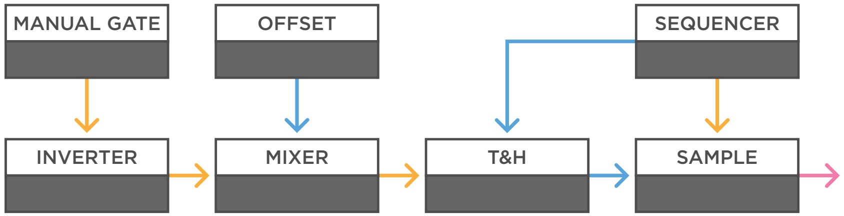
Patch-tips

Audio

Control Voltage

Trigger / Gate

MONOTRAIL
TECH TALK



(a)

a - Using a track & hold to influence a sequencer can lead to great results as well. For example, when you use a sequencer's clock to trigger a sample player and put its CV output through a track & hold module to select samples. In this setup the sequence is only passed on to the sample player when you send the track & hold a gate.
But the sequence will freeze when the gate drops out.

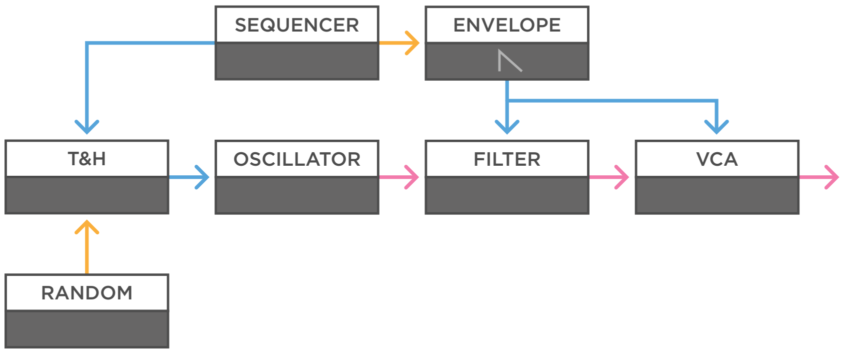
If you like to invert this behavior, and for example want to use a manual gate to freeze the sequence, you can send your manual gate to an inverter, and then mix it with offset. Now you have a gate that's always high, and will only go low, if you send a manual gate into this setup.

Track & Hold

Patch-tips

Audio
Control Voltage
Trigger / Gate

MONOTRAIL
TECH TALK



(a)

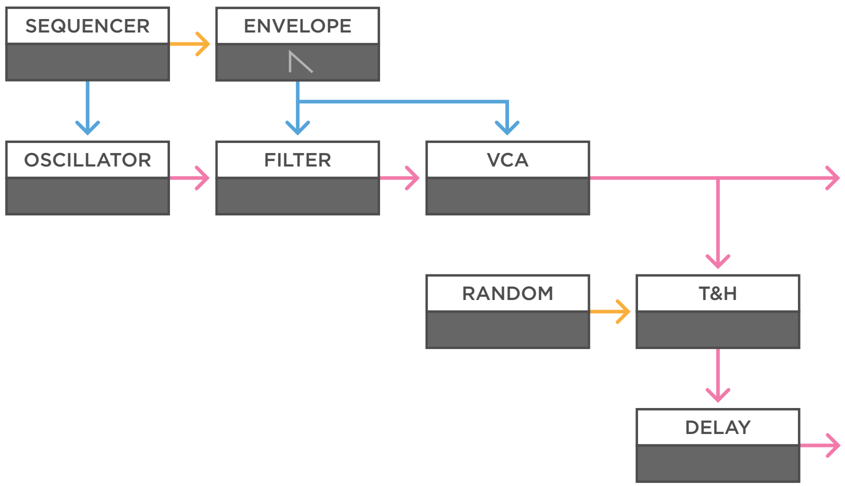
a - A similar setup with a track & hold can be used on melodic sequences. Just make a voice with an oscillator, filter, VCA, and a sequencer triggering an envelope. Use the envelope on the filter or VCA, but send the melodic sequence through the track & hold module, before going to the oscillator.

Now you can experiment with different random or clocked gates to freeze the melody for some time. Again, it can be worth it to add an inverter and/or offset voltage to an incoming signal, like a random voltage, to influence the amount of voltage that is below the tracking threshold. Some track & hold circuits are better at holding a steady value than others.

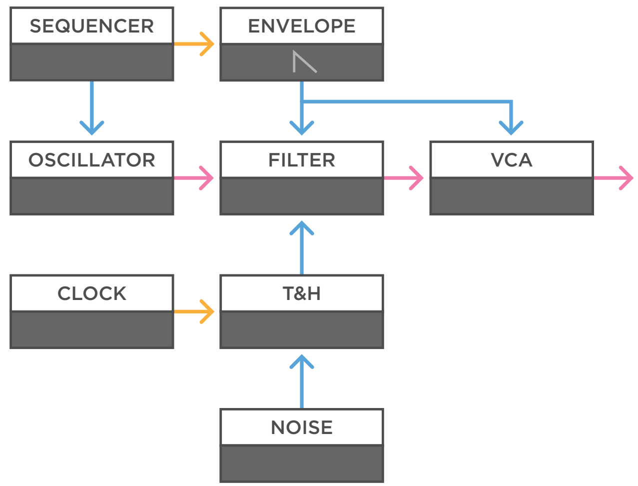
Track & Hold

Patch-tips

Audio
Control Voltage
Trigger / Gate



(a)



(b)

a - Because a track & hold module works great with audio rates, you can use it to chop up audio signals as well. For example, when you have a simple voice creating a melody, you can multiply the output signal and send one of them to a mixer, so you can just hear the voice on its own, but send the multiplication to a track & hold module. Again, send the track & hold a random voltage. Now every time the random voltage is high enough, the incoming audio is tracked, and passed on. You can send these random snippets of audio to something like a delay, or other effects modules.

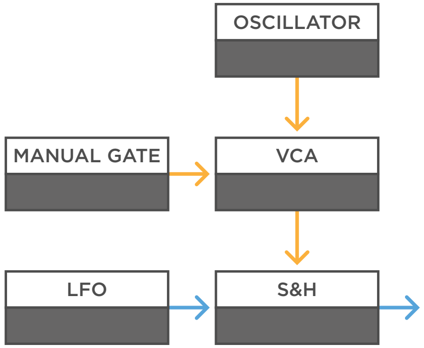
b - You can use many of the patches from the sample & hold video (Tech Talk #24), and remake them with a track & hold, to see what happens. For example, take a simple voice, and feed noise and a steady clock to a track & hold module, instead of a sample & hold module influencing the filter. Instead of a stepped random voltage, you will get a signal that alternates between a steady random voltage, and the noise influencing the filter directly. This will create interesting textures.

Track & Hold

Track & Hold from Sample & Hold

MONOTRAIL
TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

a - You can get a track & hold function from a sample & hold module, because analog sample & hold modules are really good at dealing with audio-rate signals. When you feed something like a slow sine wave LFO into the sample input, and use an audio rate gate signal, like the square wave output of a steady high pitched oscillator, into the gate input, you will create a tracking function. The module will sample & hold the incoming signal with such a high rate, that the output signal will be effectively the same as the incoming signal.

In order to add the hold function, all you have to do is send the oscillator through a closed VCA, before hitting the sample & hold. Now, when you send the VCA a gate to open it, the sample & hold module will track the incoming signal, and hold it, when you stop feeding a gate to the VCA. Just like a track & hold circuit.